Cell Biology & Bioenergetics - Lecture Notes (Module 3)

Chapter 1: Eukaryotic Cell Architecture

The eukaryotic cell is defined as a complex structural unit bounded by a plasma membrane containing membrane-bound organelles and a distinct nucleus.
Unlike prokaryotes, eukaryotic cells maintain compartmentalized biochemical microenvironments.

The nucleus refers to the membrane-enclosed organelle that houses the genetic material in the form of linear DNA and coordinates cell activities such as growth, metabolism, and protein synthesis.
Chromatin is defined as the complex of genomic DNA and histone proteins that packages genetic material inside the nuclear envelope.
The nucleolus represents the dense subnuclear region responsible for ribosomal RNA transcription and ribosome subunit assembly.

Chapter 2: Organelles and Cellular Energetics

Mitochondria are defined as double-membraned eukaryotic organelles that generate the majority of cellular adenosine triphosphate through oxidative phosphorylation.
Often referred to as the powerhouses of the cell, mitochondria possess their own circular mitochondrial DNA and ribosomes.
The mitochondrial matrix is the internal fluid compartment containing metabolic enzymes for the tricarboxylic acid cycle.

Cellular respiration refers to the metabolic pathway that breaks down glucose molecules to produce ATP in the presence of oxygen.
Glycolysis is defined as the initial anaerobic enzymatic sequence occurring in the cytosol that converts one glucose molecule into two molecules of pyruvate.
ATP synthase is the transmembrane enzyme complex that harnesses the proton electrochemical gradient across the inner mitochondrial membrane to synthesize adenosine triphosphate.

Chapter 3: The Endomembrane System

The endoplasmic reticulum is defined as an extensive interconnected network of tubular membranes and flattened cisternae continuous with the outer nuclear membrane.
Rough endoplasmic reticulum is studded with ribosomes and specializes in protein synthesis and folding for secretable proteins.
Smooth endoplasmic reticulum refers to the tubular agranular membrane network dedicated to lipid synthesis, carbohydrate metabolism, and calcium ion sequestration.

Ribosomes are defined as macromolecular ribonucleoprotein complexes that catalyze the translation of messenger RNA into polypeptide chains.
Each functional ribosome consists of a small subunit that decodes mRNA and a large subunit that forms peptide bonds.

The Golgi apparatus refers to the eukaryotic organelle composed of polar cisternae that modifies, sorts, and packages glycoproteins and lipids for secretion or delivery to lysosomes.
Lysosomes are defined as acidic, membrane-delimited digestive vesicles containing hydrolytic enzymes capable of degrading macromolecules and cellular debris.
Autophagy is the catabolic cellular process whereby damaged organelles are degraded within autophagosomes fused to lysosomes.